DO YOU KNOW?

$$\begin{pmatrix}3 & 2\\ 4 & 5\end{pmatrix}+\begin{pmatrix}1 & 6\\ 7 & 8\end{pmatrix}=\begin{pmatrix}4 & 8\\ 11 & 13\end{pmatrix}$$

$$\begin{vmatrix}3 & 2\\ 4 & 5\end{vmatrix}+\begin{vmatrix}1 & 6\\ 7 & 8\end{vmatrix}\neq\begin{vmatrix}4 & 8\\ 11 & 13\end{vmatrix}$$

Example 1.6

Evaluate $\begin{vmatrix} x & x+1\\ x-1 & x\end{vmatrix}$

Solution

$$\begin{vmatrix} x & x+1\\ x-1 & x\end{vmatrix} = x^2-(x-1)(x+1)$$
$$= x^2-(x^2-1)$$
$$= x^2-x^2+1 = 1$$

NOTE

The value of the determinant of triangular matrix is equal to the product of the main diagonal elements.

Example 1.7

Solve $\begin{vmatrix} x-1 & x & x-2\\ 0 & x-2 & x-3\\ 0 & 0 & x-3\end{vmatrix} = 0$

Solution

$$\begin{vmatrix} x-1 & x & x-2\\ 0 & x-2 & x-3\\ 0 & 0 & x-3\end{vmatrix} = 0$$

$$\Rightarrow (x-1)(x-2)(x-3)=0$$
$$x = 1, x = 2, x = 3$$

Example 1.8

Evaluate $\begin{vmatrix} 1 & 3 & 4\\ 102 & 18 & 36\\ 17 & 3 & 6\end{vmatrix}$

Solution

$$\begin{vmatrix} 1 & 3 & 4\\ 102 & 18 & 36\\ 17 & 3 & 6\end{vmatrix} = 6\begin{vmatrix} 1 & 3 & 4\\ 17 & 3 & 6\\ 17 & 3 & 6\end{vmatrix}$$
$$= 0 \quad (\text{since } R_2 \equiv R_3)$$

Example 1.9

Evaluate $\begin{vmatrix} 1 & a & a^2\\ 1 & b & b^2\\ 1 & c & c^2\end{vmatrix} = (a-b)\,(b-c)\,(c-a)$

Solution

$$\begin{vmatrix} 1 & a & a^2\\ 1 & b & b^2\\ 1 & c & c^2\end{vmatrix} = \begin{vmatrix} 0 & a-b & a^2-b^2\\ 0 & b-c & b^2-c^2\\ 1 & c & c^2\end{vmatrix} \begin{matrix} R_1 \to R_1 - R_2\\ R_2 \to R_2 - R_3\end{matrix}$$
$$= \begin{vmatrix} 0 & a-b & (a-b)(a+b)\\ 0 & b-c & (b-c)(b+c)\\ 1 & c & c^2\end{vmatrix}$$
$$= (a-b)(b-c)\begin{vmatrix} 0 & 1 & a+b\\ 0 & 1 & b+c\\ 1 & c & c^2\end{vmatrix}$$
$$= (a-b)(b-c)[0-0+$$
$$\{b+c-(a+b)\}]$$
$$= (a-b)(b-c)(c-a).$$

Exercise 1.1

1. Find the minors and cofactors of all the elements of the following determinants.

   (i) $\begin{vmatrix} 5 & 20\\ 0 & -1\end{vmatrix}$ (ii) $\begin{vmatrix} 1 & -3 & 2\\ 4 & -1 & 2\\ 3 & 5 & 2\end{vmatrix}$

2. Evaluate $\begin{vmatrix} 3 & -2 & 4\\ 2 & 0 & 1\\ 1 & 2 & 3\end{vmatrix}$

3. Solve: $\begin{vmatrix} 2 & x & 3\\ 4 & 1 & 6\\ 1 & 2 & 7\end{vmatrix} = 0$

4. Find $|AB|$ if $A = \begin{bmatrix} 3 & -1\\ 2 & 1\end{bmatrix}$ and $B = \begin{bmatrix} 3 & 0\\ 1 & -2\end{bmatrix}$

5. Solve: $\begin{vmatrix} 7 & 4 & 11\\ -3 & 5 & x\\ -x & 3 & 1\end{vmatrix} = 0$

6. Evaluate: $\begin{vmatrix} 1 & a & a^2-bc\\ 1 & b & b^2-ca\\ 1 & c & c^2-ab\end{vmatrix}$

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